

Research Profile

Computer science researcher specializing in spatial data management, graph algorithms, and constrained path-query processing in large time-dependent networks. My doctoral research at the Indian Institute of Technology Ropar focused on efficient algorithms for preference-aware routing and constrained optimization, with an emphasis on computational scalability. My broader research interests include scalable graph-query processing, parallel algorithms, and high-performance computing.

Research Interests

- **Spatial and Graph Data Management:** Efficient storage, querying, and processing of large-scale spatial, spatiotemporal, and graph data.
- **Graph Algorithms and Path Query Processing:** Constrained path queries, preference-aware routing, time-dependent networks, and scalable graph-search algorithms.
- **Parallel and High-Performance Computing:** Parallel graph algorithms, multicore computation, and GPU-accelerated approaches for large-scale graph processing.

Education

Ph.D. in Computer Science and Engineering

2021–2026

Indian Institute of Technology Ropar, Punjab, India

- **Dissertation:** “Constrained Maximization of Preferences on Time-Dependent Road Networks”
- **Ph.D. Defence:** Successfully completed on 16 September 2026.
- **Advisors:** Venkata M. V. Gunturi and Venkata Kalyan Tavva.
- **CGPA:** 7.92/10.00

M.Tech. in Computer Science and Engineering

2019–2021

Indian Institute of Technology Ropar, Punjab, India

- **Thesis:** “A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks”
- **CGPA:** 8.39/10.00

B.Tech. in Computer Science and Engineering

2014–2018

Kalyani Government Engineering College, West Bengal, India

- **CGPA:** 7.45/10.00

Research Contributions

Preference-Aware Optimization in Time-Dependent Networks

Investigated constrained preference maximization in time-dependent road networks, including parallelizable approaches for processing preference-aware path queries in large graphs.

Constrained Path Query Processing

Studied interval-based and multithreaded algorithms for constrained path optimization, addressing the computational challenges of processing path queries in large road networks.

Parallel Algorithms and System-Level Optimization

Contributed to research on parallel graph algorithms and fairness-conscious cache replacement policies for shared last-level caches in multicore systems.

Publications

Journal Publications

- [J1] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs,” *Geoinformatica*, vol. 30, article no. 19, 2026. <https://doi.org/10.1007/s10707-026-00577-z> **Impact Factor: 2.5**

Conference Publications

- [C1] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “A User-Configurable Navigation System for Wideness and Turn-Aware Routing,” in *Proceedings of the 27th International Conference on Mobile Data Management (MDM 2026)*, pp. 339–342, 2026. <https://doi.org/10.1109/MDM71479.2026.00050> **CORE Ranking: B**
- [C2] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “Interval Based Constrained Path Optimization in Time-Dependent Road Networks,” in *Web Information Systems Engineering – WISE 2025*, pp. 1–12, 2025. https://doi.org/10.1007/978-981-95-7251-9_12 **CORE Ranking: B**
- [C3] **Kousik Kumar Dutta**, Ankita Dewan, and Venkata M. V. Gunturi, “A Multi-Threading Algorithm for Constrained Path Optimization Problem on Road Networks,” in *Web Information Systems Engineering – WISE 2022*, pp. 110–118, 2022. https://doi.org/10.1007/978-3-031-20891-1_9 **CORE Ranking: B**
- [C4] **Kousik Kumar Dutta**, Prathamesh Nitin Tanksale, and Shirshendu Das, “A Fairness Conscious Cache Replacement Policy for Last Level Cache,” in *Proceedings of the Design, Automation & Test in Europe Conference & Exhibition (DATE 2021)*, pp. 695–700, 2021. <https://doi.org/10.23919/DATE51398.2021.9474096> **CORE Ranking: A**
- [C5] Ankita Dewan, Venkata M. V. Gunturi, Vinayak Naik, and **Kousik Kumar Dutta**, “NEAT Activity Detection Using Smartwatch at Low Sampling Frequency,” in *IEEE SmartWorld/SCALCOM/UIC/ATC/IOP/SCI 2021*, pp. 25–32, 2021. <https://doi.org/10.1109/SWC50871.2021.00014> **CORE Ranking: B**

Symposium and Workshop Publications

- [W1] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “Constrain Path Optimization on Time-Dependent Road Networks,” in *Web Information Systems Engineering – WISE 2024 PhD Symposium, Demos and Workshops*, pp. 274–282, 2024. https://doi.org/10.1007/978-981-96-1483-7_24 **CORE Ranking: B**

Extended Abstracts

- [E1] Prathamesh Nitin Tanksale, **Kousik Kumar Dutta**, Shirshendu Das, and T. V. Kalyan, “CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC,” extended abstract, in *IEEE International Conference on High Performance Computing (HiPC 2023) Workshops*, 2023. <https://doi.org/10.1109/HiPCW61695.2023.00021>

Manuscripts Under Review

- [U1] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “A Bi-Directional Search for Interval Constrained Path Optimization in Large Time-Dependent Road Networks,” submitted to *Knowledge and Information Systems*, 2026 (major revision submitted).
- [U2] **Kousik Kumar Dutta** and Venkata M. V. Gunturi, “A Decomposition-Based Heuristic for Selective Pickup–Delivery Routing with Non-Additive Loading–Unloading Costs,” submitted to *Annals of Operations Research*, 2026.

Teaching Experience

Teaching Assistant

Indian Institute of Technology Ropar

Databases (B.Tech. Core Course)

7 Offerings, 2021–2025

- Conducted laboratory sessions and evaluated assignments on SQL, relational database design, and query processing.
- Mentored student teams in database-driven application development and course projects.

Software Engineering (B.Tech. Core Course)

5 Offerings, 2020–2025

- Evaluated laboratory assignments and supported instruction in software development practices.
- Guided students in requirements analysis, software design, testing, and project implementation.

Postgraduate Software Lab (M.Tech. Core Course)

1 Offering, 2020

- Mentored postgraduate students in advanced data structures and algorithm implementation.

Teaching Interests: Database Management Systems, Data Structures, Spatial Computing and Advanced Data Structures.

Academic Service

- **Reviewer, Geoinformatica (2023–Present):** Completed peer reviews of three manuscripts in areas including spatial databases, graph processing, and route planning.
- **Technical Committee Head, SYNTACS 2026:** Symposium on Novel Technologies and Advances in Computer Science, Department of Computer Science and Engineering, Indian Institute of Technology Ropar.

Awards, Fellowships and Grants

- **IEEE TCDE Student Grant (2026):** Recipient of the IEEE Technical Committee on Data Engineering Student Grant for participation in MDM 2026.
- **Institute Fellowship (Ph.D.):** Full doctoral fellowship, Indian Institute of Technology Ropar, 2021–2026.
- **MHRD Scholarship (M.Tech.):** Ministry of Human Resource Development scholarship for postgraduate studies at IIT Ropar, 2019–2021.
- **GATE 2019:** All India Rank 644 in Computer Science and Engineering.

Selected Conference Presentations

- **MDM 2026:** “A User-Configurable Navigation System for Wideness and Turn-Aware Routing.”
- **WISE 2025:** “Interval Based Constrained Path Optimization in Time-Dependent Road Networks.”
- **WISE 2024 PhD Symposium:** “Constrained Path Optimization on Time-Dependent Road Networks.”

Technical Expertise

- **Programming:** JAVA, C, C++, Python.
- **Parallel and Research Computing:** Multithreading, OpenMP, Linux.
- **Development and Research Tools:** Git, L^AT_EX.

References

Available upon request.